Audio & Speech: Environment Sound Classification

EE366200: Digital Signal Processing Lab

Lab3 – Sep. 22, 2025

Prof. Chi-Chun Lee

TAs: 莊翔凱、周星航

BIIC

Sound Classification Flow Chart

Read Data → Feature Extraction → Cross-Validation → Testing & Evaluation

Read Data

Librosa
Soundfile

Feature Extraction

Librosa
Essentia
Opensmile

Cross-Validation

Scikit-Learn

Testing & Evaluation

Scikit-Learn

Data

- The ESC dataset is a collection of short environmental recordings available in a unified format (5-second-long clips, 44.1 kHz, single channel)
- Total: 400 Files, 40 in each class

Karol J. Piczak, 2015, "ESC: Dataset for Environmental Sound Classification"

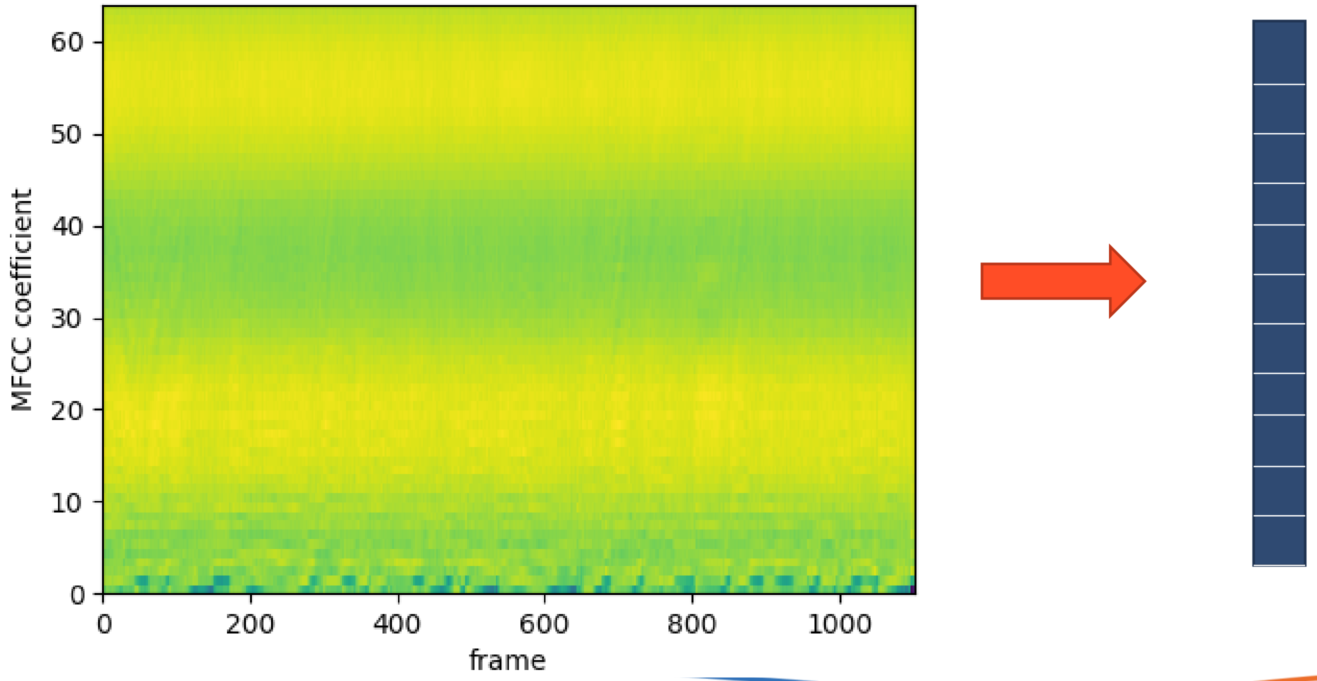
Dog bark	Rain	Sea waves	Baby cry	Clock tick
Person sneeze	Helicopter	Chainsaw	Rooster	Fire crackling

Component (1) – Acoustic Feature Extraction

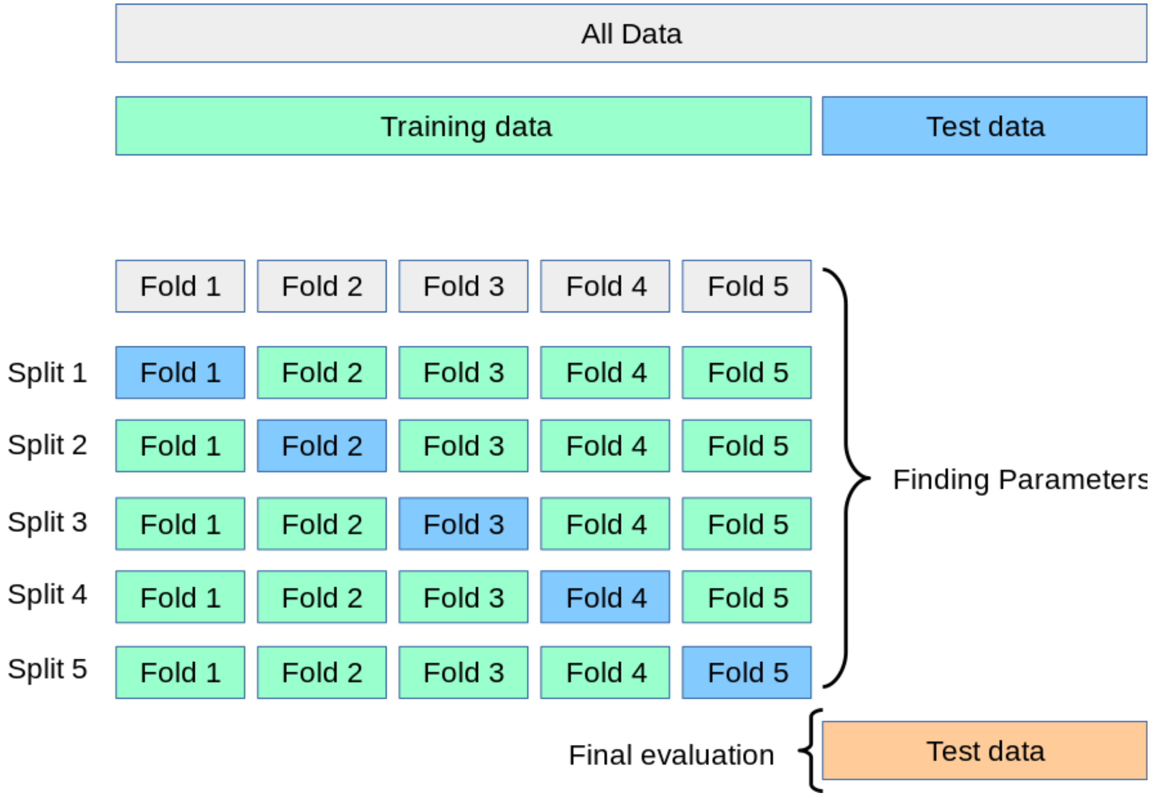
Use librosa.load to read sound files

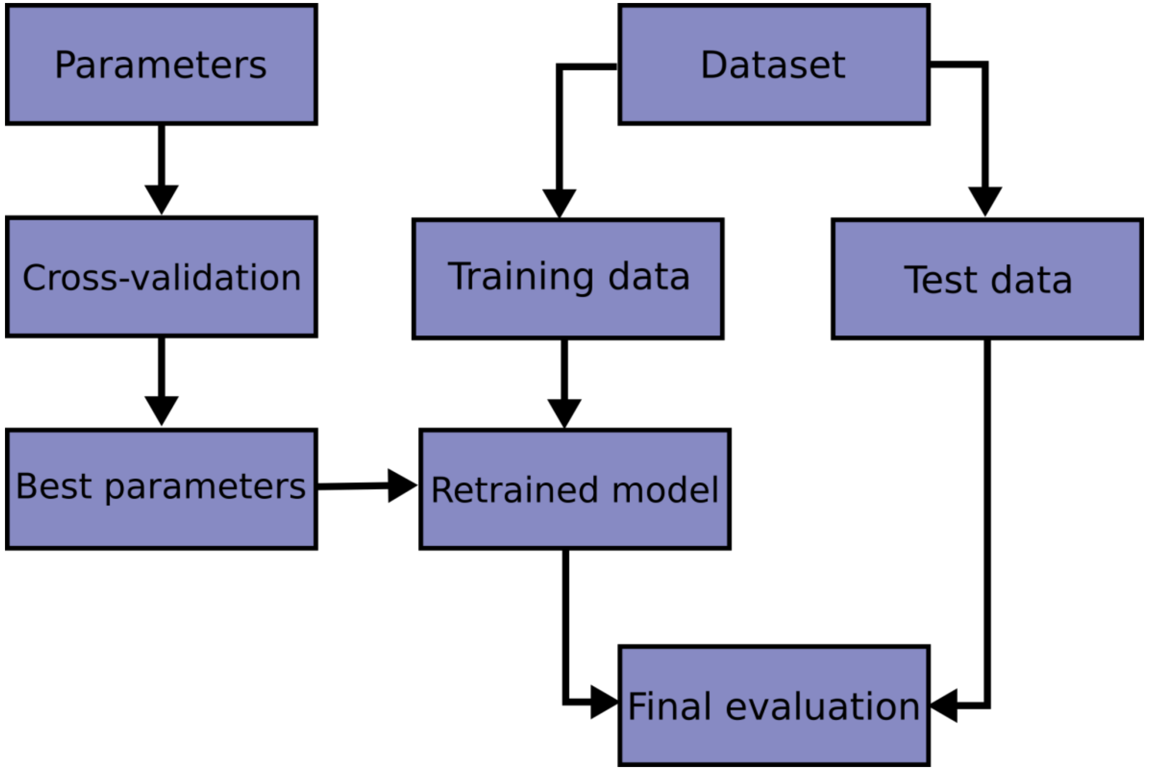
Use librosa.feature.mfcc to calculate MFCC (returns 2D N*T array)

Get statistic (mean, std, …) along time axis to reduce it into 1D vector

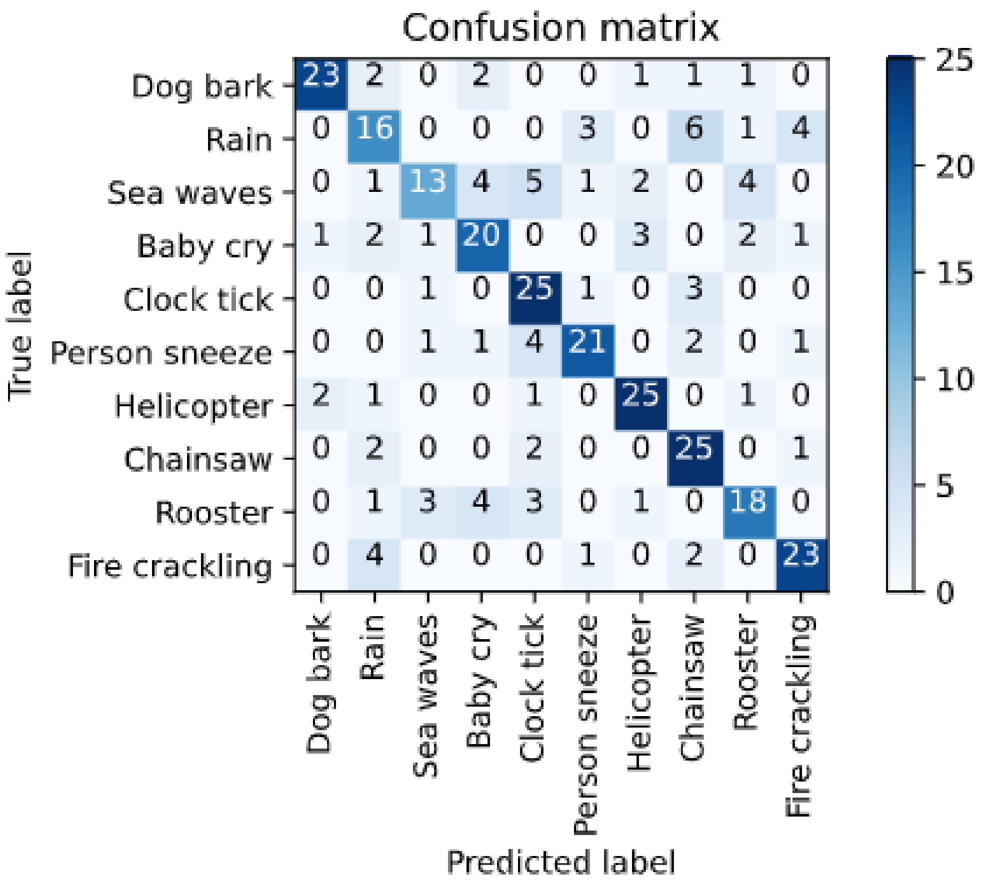


Component (2) – Cross-Validation



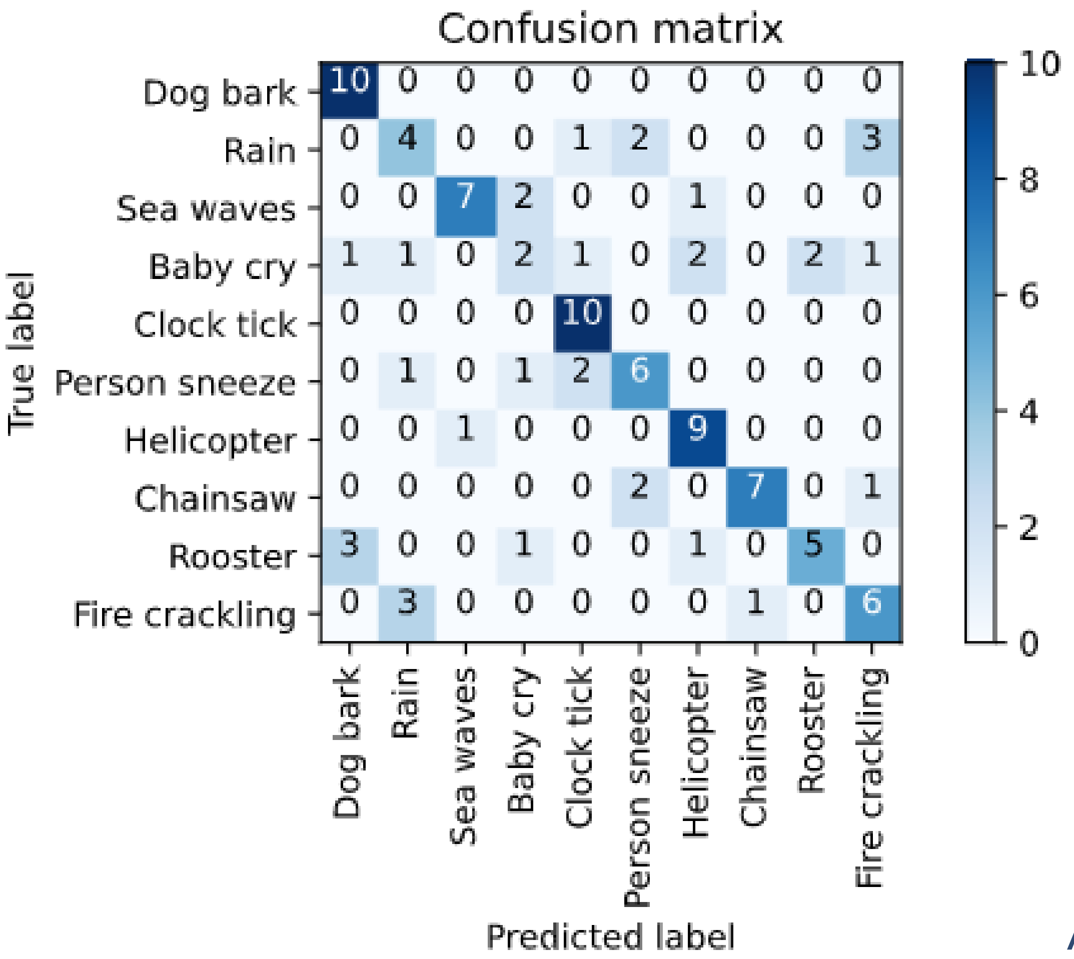


Demo (1) – Cross Validation Results



ACC = 0.697

Demo (2) – Testing Evaluation



	Dog bark	Rain	Sea waves	Baby cry	Clock tick	Person sneeze	Helicopter	Chainsaw	Rooster	Fire crackling
Dog bark	10	0	0	0	0	0	0	0	0	0
Rain	0	4	0	0	1	2	0	0	0	3
Sea waves	0	0	7	2	0	0	1	0	0	0
Baby cry	1	1	0	2	1	0	2	0	2	1
Clock tick	0	0	0	0	10	0	0	0	0	0
Person sneeze	0	1	0	1	2	6	0	0	0	0
Helicopter	0	0	1	0	0	0	9	0	0	0
Chainsaw	0	0	0	0	0	2	0	7	0	1
Rooster	3	0	0	1	0	0	1	0	5	0
Fire crackling	0	3	0	0	0	0	0	1	0	6

ACC = 0.66

Report Questions

- (Bonus) Try different features, statistics functions, classifiers to achieve higher performance

- (Bonus) You get extra credit if you achieve 75+% accuracy on testing set. Briefly explain what and how you managed to make the prediction better.